

SEMESTER-III

UES009: MECHANICS				
	L	T	P	Cr
	2	1	2 [#]	2.5
	(#: Two hours Lab once in a semester)			
Course Objective: The objective of this module is to help students develop the techniques needed to solve general engineering mechanics problems. Students will learn to describe physical systems mathematically so that their behavior can be predicted.				
Syllabus				
Review of Newton’s law of motion and vector algebra.				
Equilibrium of Bodies: Free-body diagrams; conditions of equilibrium; torque due to a force; statical determinacy.				
Plane Trusses: Forces in members of a truss by method of joints and method of sections.				
Friction: Sliding, belt, screw and rolling.				
Properties of Plane Surfaces: First moment of area; centroid; second moment of area etc.				
Shear Force and Bending Moment Diagrams: Types of load on beams; classification of beams; axial, shear force and bending moment diagrams: simply supported, overhung and cantilever beams subjected to any combination of point loads, uniformly distributed and varying load and moment.				
Virtual Work: Principle of virtual work; calculation of virtual displacement and virtual work.				
Laboratory Work				
Experimental Project Assignment/ Micro Project: Students in groups of 4/5 will do project on Model Bridge Experiment: This will involve construction of a model bridge using steel wire and wood.				
Course Learning Outcomes (CLOs)				
The students will be able to:				
1. determine resultants in plane force systems				
2. identify and quantify all forces associated with a static framework				
3. draw Shear Force Diagram and Bending Moment Diagram in various kinds of beams subjected to different kinds of loads				
Text Books				
1. Shames, I. H. Engineering Mechanics: Dynamics, Pearson Education India (2006).				
2. Beer, Johnston, Clausen and Staab, Vector Mechanics for Engineers, Dynamics, McGraw- Hill Higher Education (2003).				
Reference Books				
1. Hibler, T.A., Engineering Mechanics: Statics and Dynamics, Prentice Hall (2012).				
2. Timoshenko and Young, Engineering Mechanics, Tata McGraw Hill Education Private Limited, (2006).				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weights (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include Assignments/Projects/Tutorials/Quiz	25

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